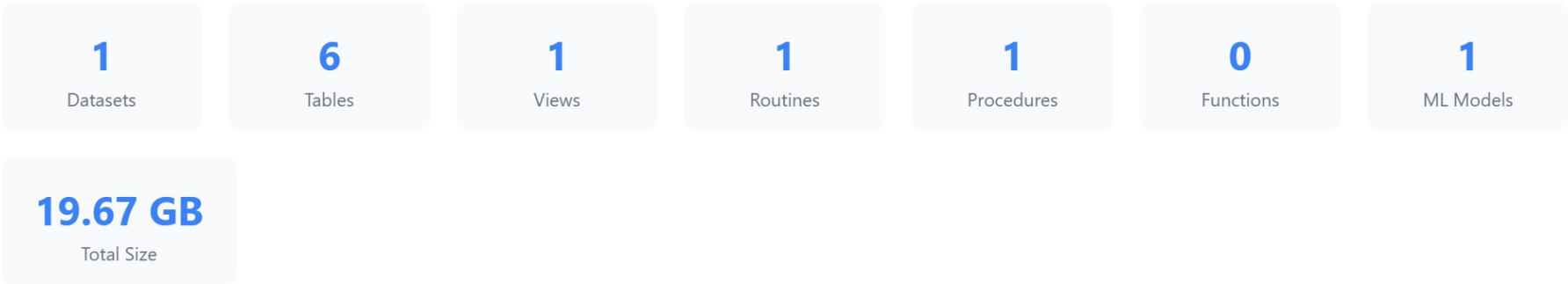


assess

Assessment Report • Generated 9/3/2026, 5:50:55 pm IST
Project: bigquery • Status: completed • Datasets: 1 • Tables: 6 • Views: 1

Summary



Started: 28/2/2026, 12:18:57 pm
Completed: 28/2/2026, 12:19:29 pm

Datasets (1)

DATASET	TABLES	SIZE	LOCATION
sales_analytics	7	19.67 GB	asia-south1

Tables (7)

TABLE	TYPE	ROWS	SIZE	PARTITIONING	CLUSTERING
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sales_analytics.assess_data	BASE TABLE	30	0.00 MB	None	None
sales_analytics.assess_tbl	BASE TABLE	1,00,00,000	9.83 GB	None	None
sales_analytics.assess_tbl_part_clust	BASE TABLE	1,00,00,000	9.83 GB	created_at	id
sales_analytics.customers	BASE TABLE	3	0.00 MB	None	None
sales_analytics.orders	BASE TABLE	4	0.00 MB	None	None
sales_analytics.orders_data	BASE TABLE	4	0.00 MB	None	None
sales_analytics.temp_view	VIEW	0	0.00 MB	None	None

Views (1)

VIEW	DEFINITION
sales_analytics.temp_view	<pre>select * from sales_analytics.customers</pre>

Stored Procedures (1)

PROCEDURE	LANGUAGE	CREATED
sales_analytics.insert_orders	SQL	18/1/2026, 10:54:17 pm

ML & Spark Models (1)

MODEL	TYPE	CREATED
sales_analytics.sales_analytics.order_amount_prediction_model	LINEAR_REGRESSION	18/1/2026, 11:19:05 pm

Recommendations

Recommended: Redshift Serverless

- Low compute usage (0.2 slot-hours/month) — pay-per-use is more efficient
- Low query frequency (~32/day) — avoid paying for idle cluster
- Serverless auto-scales RPU based on workload complexity
- No cluster management overhead

Provisioned Config

Node: dc2.large × 1 | vCPU: 2 | Memory: 15 GB

Distribution & Sort Keys

TABLE	DIST KEY	SORT KEY
sales_analytics.assess_data	EVEN	created_at
sales_analytics.assess_tbl	id	created_at
sales_analytics.assess_tbl_part_clust	id	created_at, id
sales_analytics.customers	EVEN	created_at
sales_analytics.orders	EVEN	order_date
sales_analytics.orders_data	EVEN	AUTO

Architecture Strategies

Data Storage & Query Strategy

- 94.2% BI/Scheduled queries — Load all frequently queried data into Redshift for consistent, low-latency performance.
- Use Redshift materialized views for complex aggregations used by BI tools.
- Implement Redshift workload management (WLM) to prioritize BI queries.

TCO Analysis (US East (N. Virginia))

BigQuery Monthly	Provisioned Monthly	Serverless Monthly
\$0.5	\$182.5	\$0.51

3-Year TCO Comparison

BigQuery: \$18.09 | Provisioned: \$6,572.36 | Serverless: \$20.57

Best Option: serverless (Save \$-2.48 / -13.7%)

Provisioned: dc2.large × 1 nodes